Gabriel P. Oliveira

Data Scientist and Researcher

gabrielpoliveira@dcc.ufmg.br
dcc.ufmg.br/~gabrielpoliveira

Education

2021–2025 **Ph.D. in Computer Science**, *Universidade Federal de Minas Gerais (UFMG)*, Brazil.
Research stay at **Politecnico di Torino, Italy** (2024-25)
Thesis: On the dynamics of music virality and its relation with mainstream success
Advisor: Prof. Mirella M. Moro, Ph.D. **GPA:** 94.9 out of 100
Co-advisor: Prof. Ana Paula Couto da Silva, Ph.D.

2019–2021 **M.Sc. in Computer Science**, *Universidade Federal de Minas Gerais (UFMG)*, Brazil.
Thesis: Analyses of musical success based on time, genre and collaboration **GPA:** 92.0 out of 100
Advisor: Prof. Mirella M. Moro, Ph.D.
Co-advisor: Prof. Anisio Lacerda, Ph.D.

2015–2018 **B.Sc. in Computer Science**, *Universidade Federal de Minas Gerais (UFMG)*, Brazil.
GPA: 91.3 out of 100

2012–2014 **High School Technical Degree in Computer Networks**, *Centro Federal de Educação Tecnológica de Minas Gerais (CEFET-MG)*, Brazil.

Professional Experience

2024–2025 **Visiting Researcher**, *Politecnico di Torino*, Italy.
As a visitor at the SmartData@PoliTO center and the TNG Group, I investigated the viral spread of music on social platforms by modeling it as a contagion process. Building on classical epidemiological models, I developed an approach to effectively capture and interpret the dynamics underlying musical virality.

2021–2024 **Data Scientist**, *Analytical Capabilities Project*, UFMG/MPMG.
Member of the Analytical Capabilities Project, a collaboration between the Computer Science Department of UFMG and the Prosecution Service of the State of Minas Gerais (MPMG). My activities included building models for identifying fraud in public bidding processes and using machine learning and natural language processing techniques to classify and enhance public expenditure data.

2019–2023 **Research Assistant**, *Project Bàde*, UFMG.
The goal of this project was to develop semantic metrics derived from social network data to enhance several applications and services. Specifically, my research used social network analysis and data mining techniques to analyze different aspects of musical success and how they relate to collaboration and music genres.

2019 **Data Science Intern**, UFMG/Localiza.
As a member of the Data Science Residency Program, a collaboration between the Computer Science Department of UFMG and Localiza, I engaged in developing machine learning models to address practical challenges within the car rental industry. My responsibilities involved optimizing marketing operations within this sector by using techniques such as clustering and association rule mining.

2018–2019 **Software Engineering Intern**, dti digital crafters.
In this role, I focused on modeling SQL databases, crafting APIs using C# .NET, and designing interfaces with HTML5/CSS3 and AngularJS.

2017–2018 **Undergraduate Research Assistant**, *Project FoRGit*, UFMG.
The goals of this project included building a heterogeneous social collaboration network from GitHub, analyzing new parameters that measure the interactions between GitHub developers, and proposing new metrics for their social capital.

2016–2017 **Undergraduate Teaching Assistant**, *Computer Science Department*, UFMG.
Teaching assistant in the Computer Programming course (DCC001) for Engineering students.

2015 **IT Support Intern**, *Support Center for e-learning*, UFMG.
My activities included network and server management, user support, support to audiovisual content production, and web conferencing management.

Skills

Technologies Python, SQL, Jupyter, PySpark, C/C++, C#, JavaScript, Apache Hive, Git

Libraries Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, NetworkX

Hard Skills Network Science, Machine Learning, Data Mining, Statistical Analysis

Soft Skills Communication, Teamwork, Problem Solving, Career-focused

Awards

2023 **Best Master Thesis Award**, Brazilian Symposium on Databases (SBBD).
Prize awarded in the 5th Contest of Theses and Dissertations on Databases.

2022 **Best Master Thesis Award – Finalist**, Congress of the Brazilian Computer Society (CSBC).
One of the ten Master theses selected for presentation at the Theses and Dissertations Contest of the Congress of the Brazilian Computer Society.

2018 **Best Graduating Student Award – 1st Runner-up**, Universidade Federal de Minas Gerais.
Prize awarded for students with the highest GPA in the Computer Science Undergraduate Program.

Selected Publications

The complete list of publications is available on my webpage.

2024 **Analyzing the temporal relation between virality and success in the Brazilian music market**, *Integrated Software and Hardware Seminar (SEMISH)*, Brasília, Brazil, **Best Paper Award**.
GP Oliveira, APC da Silva, MM Moro

2024 **What makes a viral song? Unraveling music virality factors**, *ACM Web Science Conference (WebSci)*, Stuttgart, Germany.
GP Oliveira, APC da Silva, MM Moro

2023 **Hot streaks in the music industry: identifying and characterizing above-average success periods in artists' careers**, *Scientometrics*.
GP Oliveira, MO Silva, DB Seufitelli, GRG Barbosa, BC Melo, MM Moro

2023 **Hit song science: a comprehensive survey and research directions**, *J. of New Music Research*.
DB Seufitelli, **GP Oliveira**, MO Silva, C Scofield, MM Moro

2023 **How do developers collaborate? Investigating GitHub heterogeneous networks**, *Software Quality Journal*.
GP Oliveira, AFC Moura, NA Batista, MA Brandão, A Hora, MM Moro

2020 **Detecting Collaboration Profiles in Success-based Music Genre Networks**, *International Society for Music Information Retrieval Conference (ISMIR)*, Montréal, Canada, **Best Paper Presentation Award**.
GP Oliveira, MO Silva, DB Seufitelli, A Lacerda, MM Moro

Languages

Portuguese **Native**

English **Advanced**

Italian **Conversational/Intermediate**

Spanish **Basic**

Other activities

2025 External Reviewer – BraSNAM, WEI

2024 External Reviewer – SEMISH, DSW

2023 External Reviewer – BraSNAM, SEMISH, DSW

2022 External Reviewer – BraSNAM, SBBD

2021 External Reviewer – WebMedia, SEMISH, WEI